

NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL

WARANGAL - 506 004, TELANGANA STATE - INDIA



PROVISIONAL CERTIFICATE DOCTOR OF PHILOSOPHY (Ph.D.)

No. : **29204**

Certified that

Mr. Monex Sharma

S/O

Manisana Sharma

Roll No

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Department of

MECHANICAL ENGINEERING

has successfully completed all prescribed requirements for the award of the degree of **Doctor of Philosophy (Ph.D.)** of the institute.

The date of viva-voce examination held on: **26-05-2026**

The title of the thesis submitted is **"Terrain-Adaptive Multi-Objective Coverage Path Planning for Mobile Robots Using an Improved NSGA-II Algorithm"**.

Warangal

Dated : 08-06-2026


DEAN (ACADEMIC)



Note: Validity of this certificate will expire on the award of the degree in the next convocation.